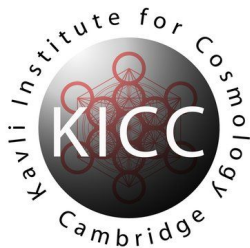


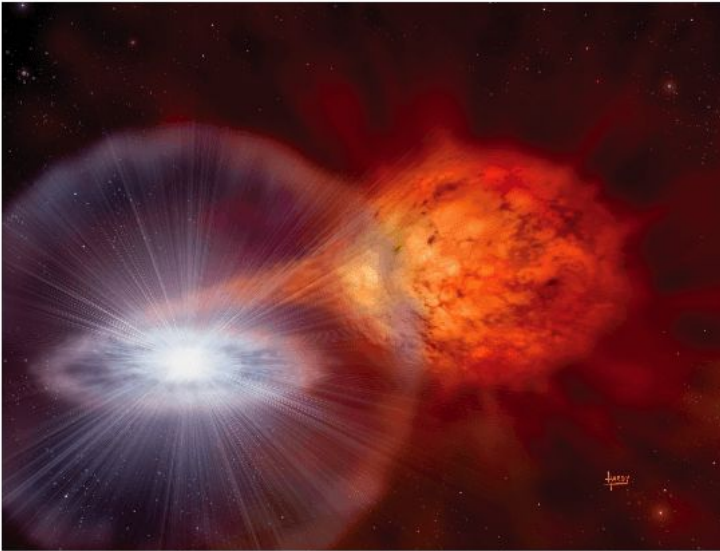
The environmental dependence of SN Ia i-band secondary maximum

Matt Grayling



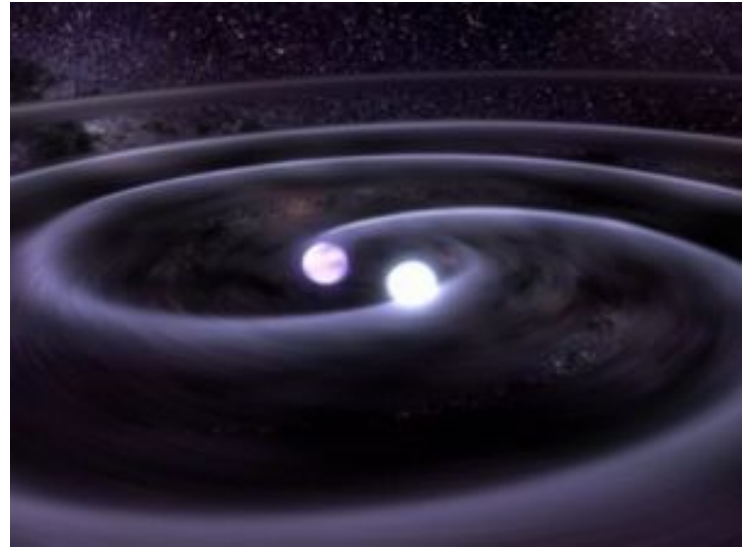
Type Ia Supernovae

Single Degenerate



Credit: David A. Hardy, © David A. Hardy/astroasrt.org

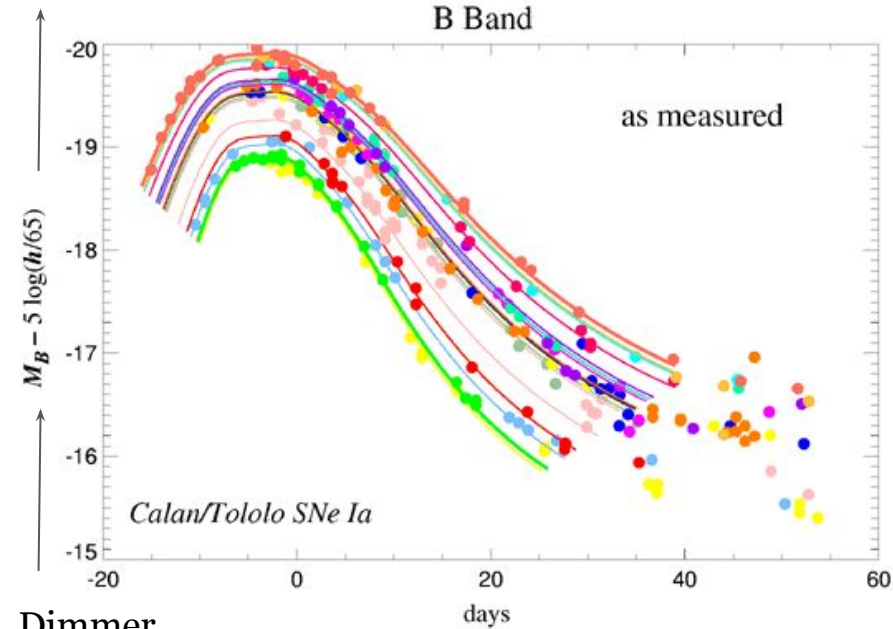
Double Degenerate



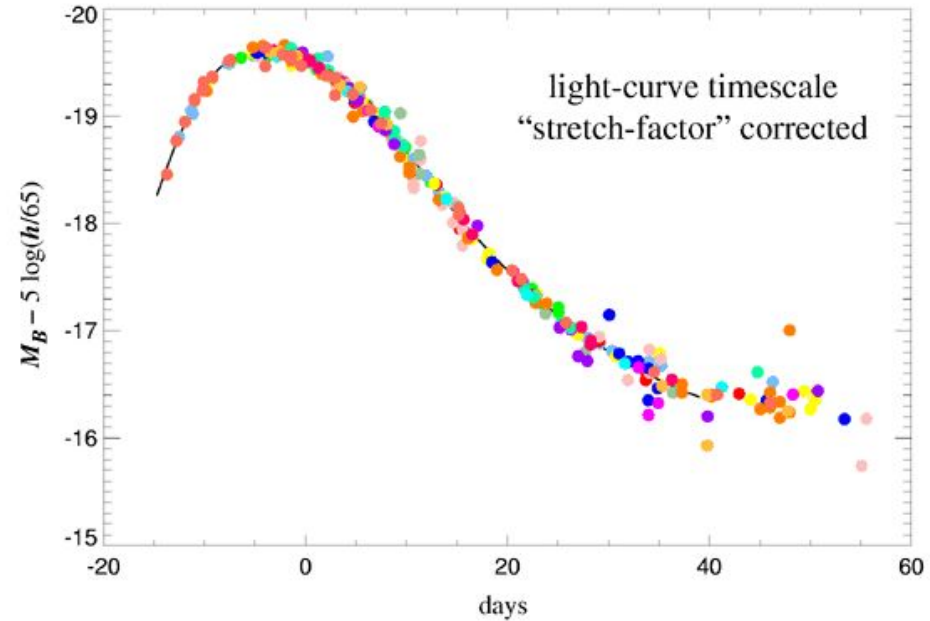
Copyright: GSFC/D. Berry.

Standard(isable) Candles

Brighter



Dimmer

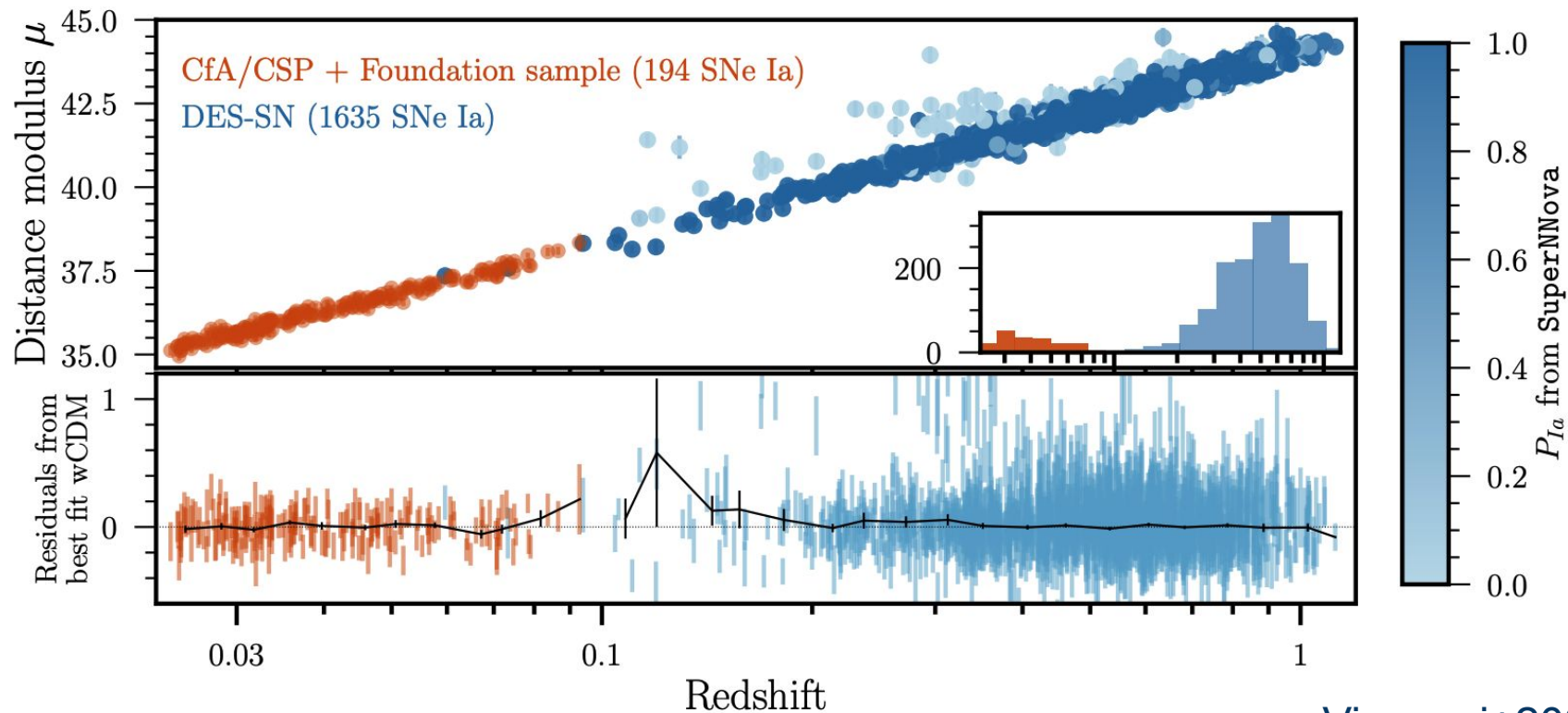


From Supernova Cosmology Project

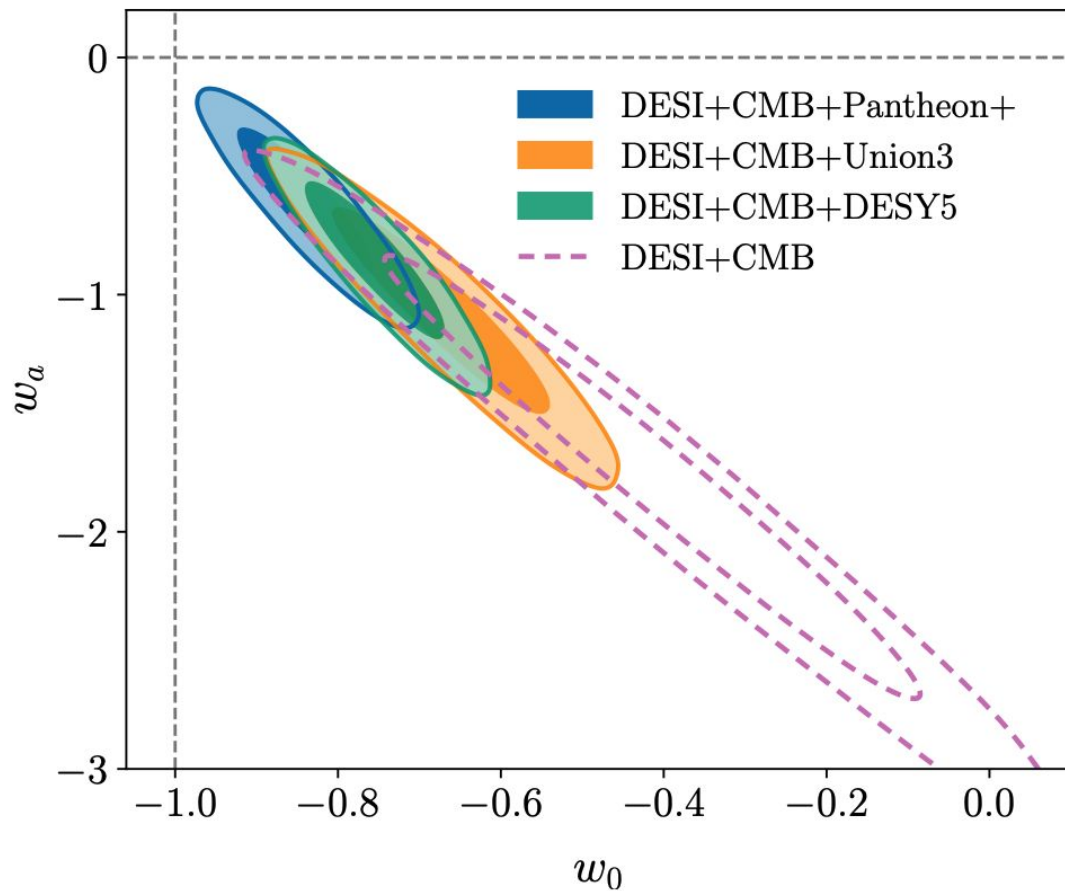
Calculating Distance

$$\mu^s = m_B^s - M + \alpha x_1^s - \beta c^s$$

The Hubble Diagram



Crisis in Cosmology?

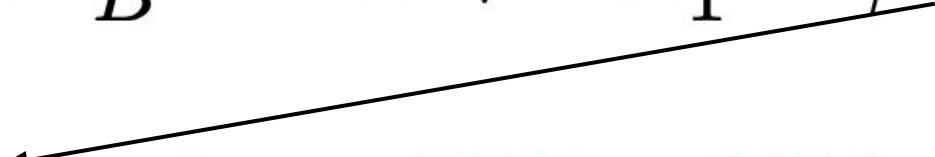


DESI Collaboration
2025

Supernova cosmology is inseparably
linked to supernova astrophysics

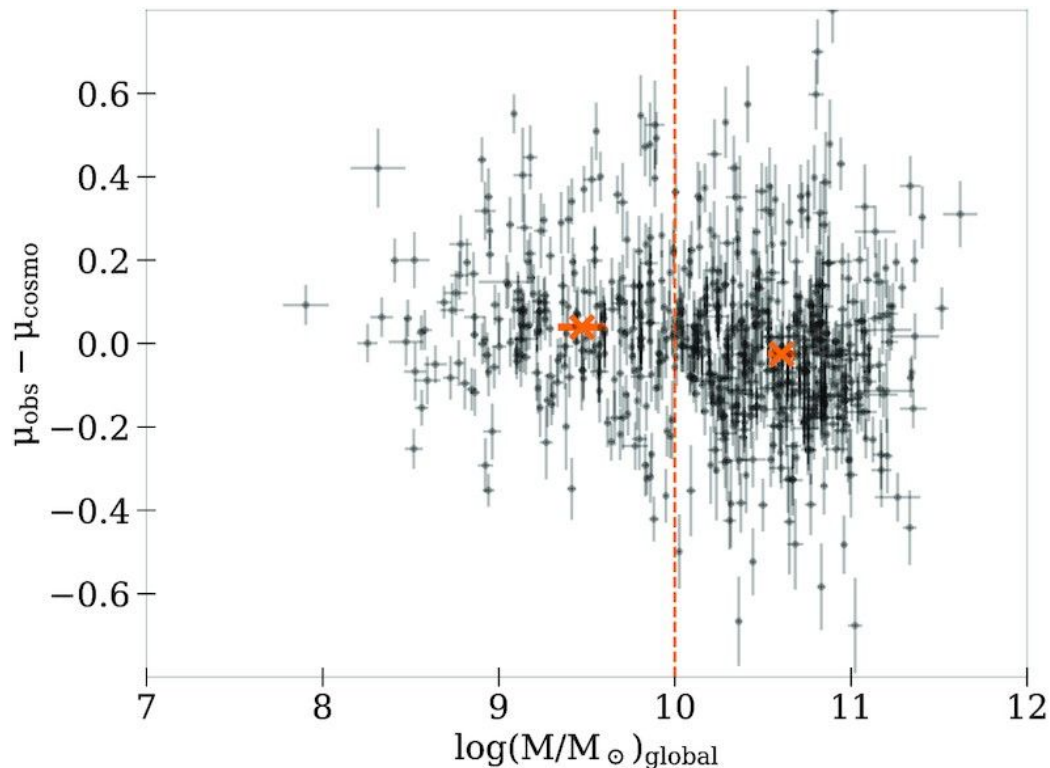
The Impact of Dust

$$\mu^s = m_B^s - M + \alpha x_1^s - \beta c^s$$

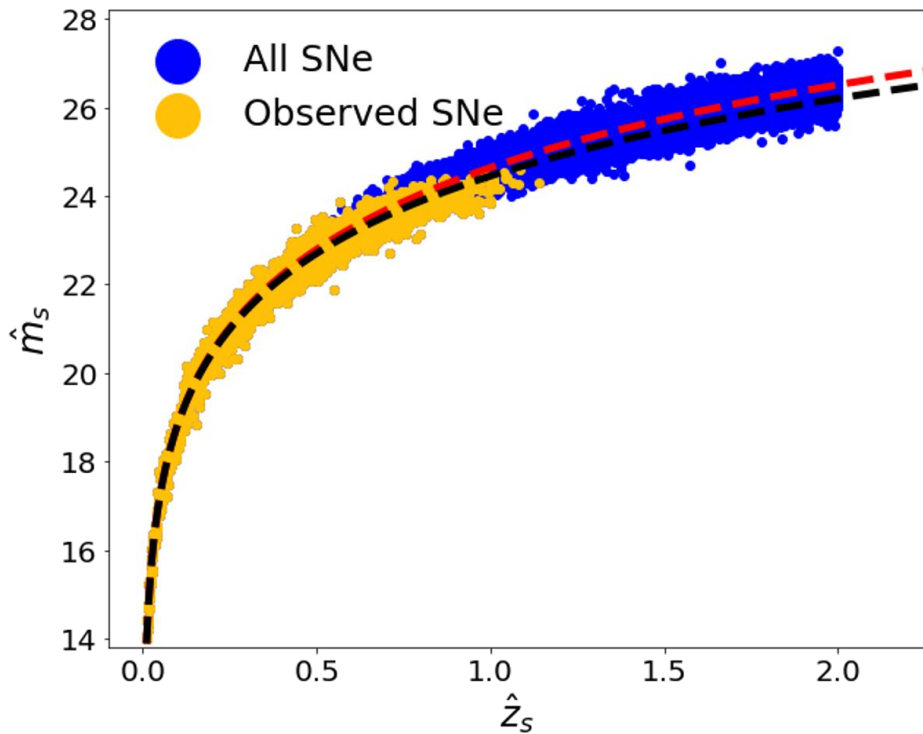
$$c^s = c_{\text{int}}^s + E(B - V)^s$$


The Mass Step

- SN Ia properties depend on environment
- Just different dust properties?
 - SN populations identical, just appear different?
- Or intrinsic differences?
 - Interesting physical implications, different progenitor channels?



Correcting for Selection Effects



- We typically correct for selection effects through simulations
- To do this, we need a forward model for the population of SNe Ia which captures these effects

Challenges in Supernova Cosmology

The majority of systematic uncertainty in DES5YR relates to unknown astrophysics

Systematic w uncertainty budget

DES-SN5YR, no CMB prior

$\sigma_w^{\text{SN-only (Total): 0.152}$

$\sigma_w^{\text{SN-only (stat) 0.132}$

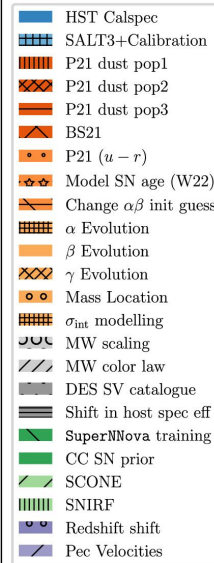
$\sigma_w^{\text{SN-only (syst-only) 0.076}$

DES-SN5YR + CMB prior

$\sigma_w^{\text{SN+CMB (Total): 0.036}$

$\sigma_w^{\text{SN+CMB (stat) 0.032}$

$\sigma_w^{\text{SN+CMB (syst-only) 0.017}$



Recipe for a Type Ia Supernova Model

Type Ia SN SED

=

+

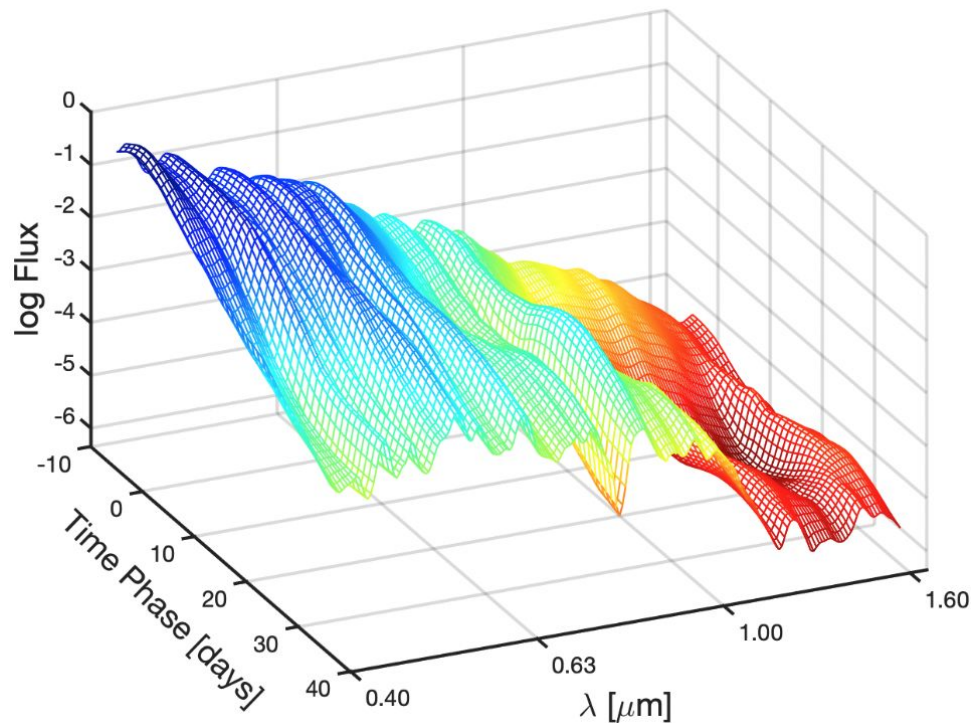
+

+



Recipe for a Type Ia Supernova Model

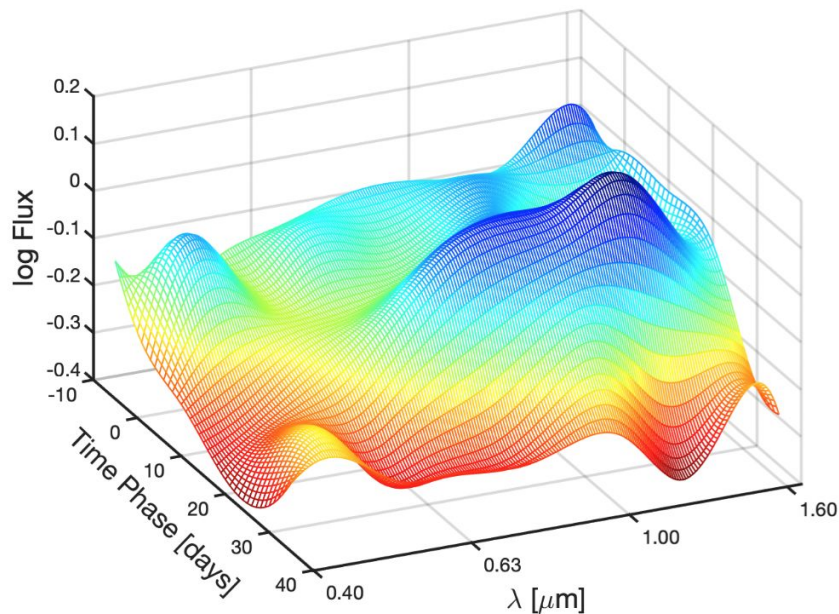
Type Ia SN SED
=
Mean Intrinsic SED
+
+
+



Recipe for a Type Ia Supernova Model

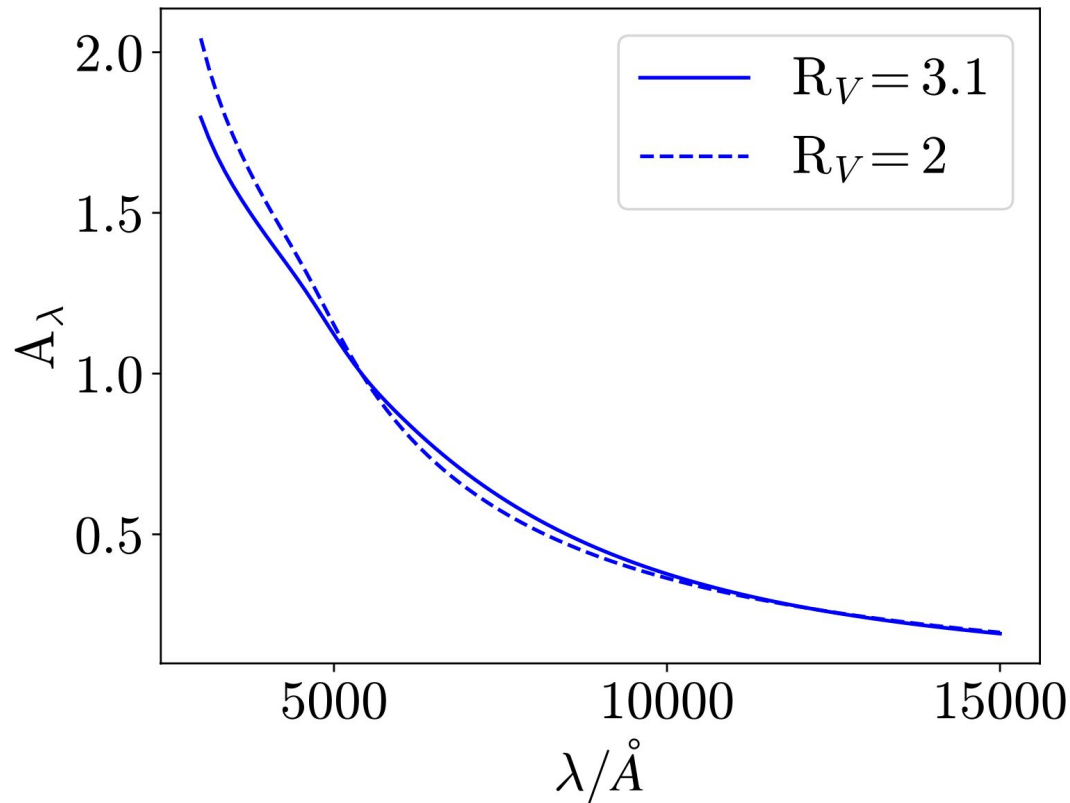
Type Ia SN SED
=
Mean Intrinsic SED
+
Effect of Stretch
+
+

$\theta_1 \times$



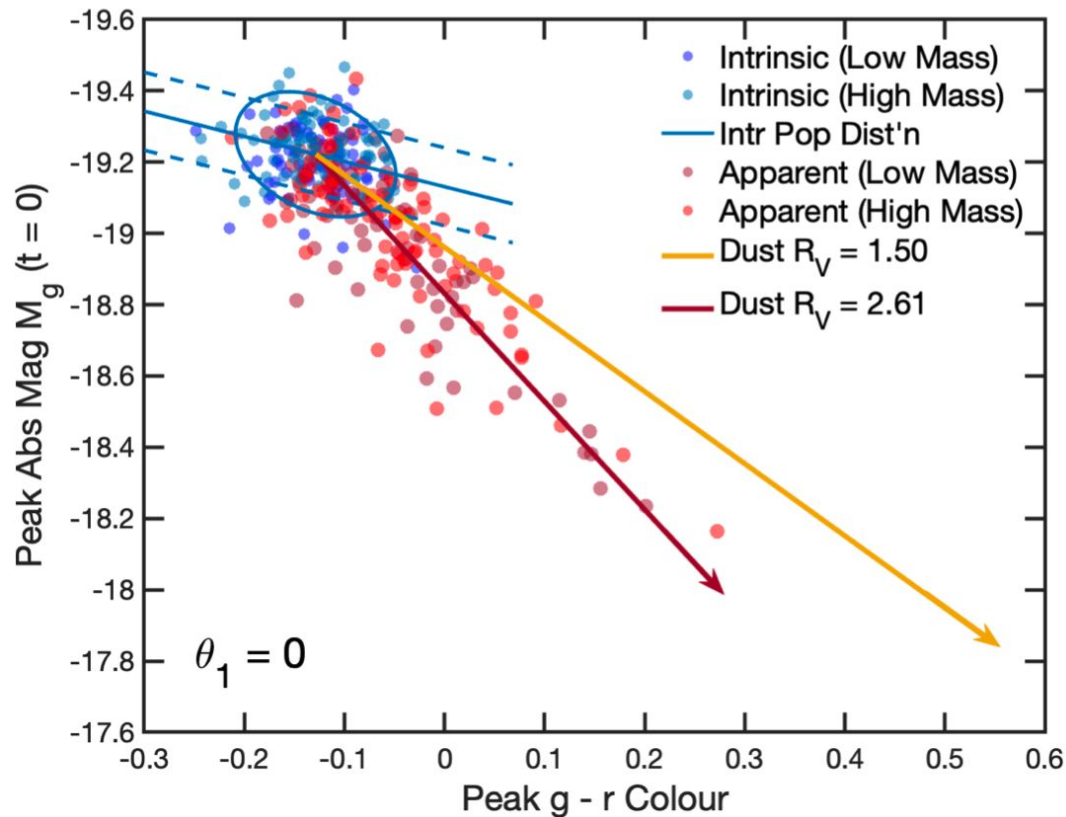
Recipe for a Type Ia Supernova Model

Type Ia SN SED
=
Mean Intrinsic SED
+
Effect of Stretch
+
Dust Extinction
+

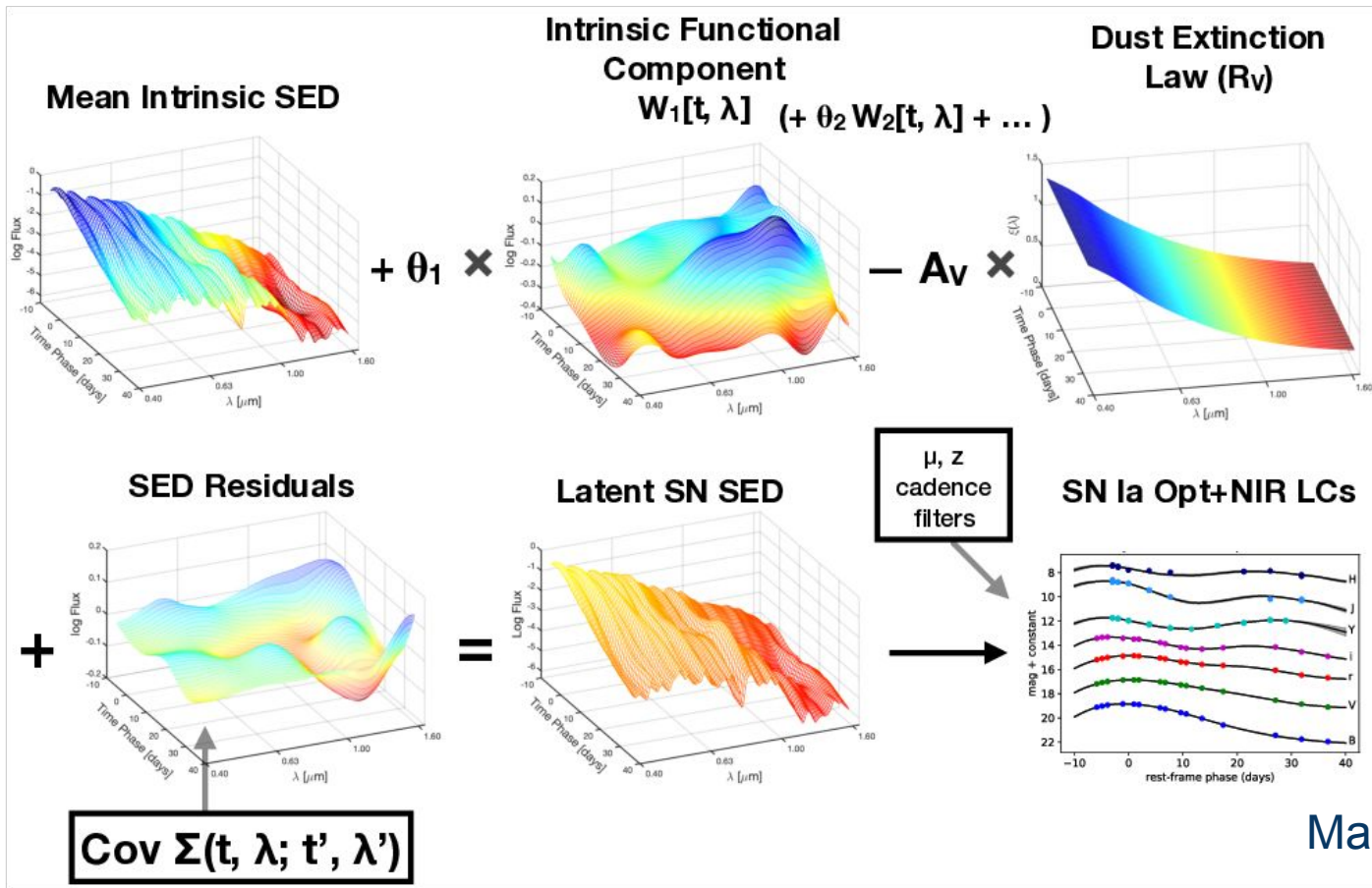


Recipe for a Type Ia Supernova Model

Type Ia SN SED
=
Mean Intrinsic SED
+
Effect of Stretch
+
Dust Extinction
+
Intrinsic Colour



The BayeSN Model



Advantages of BayeSN

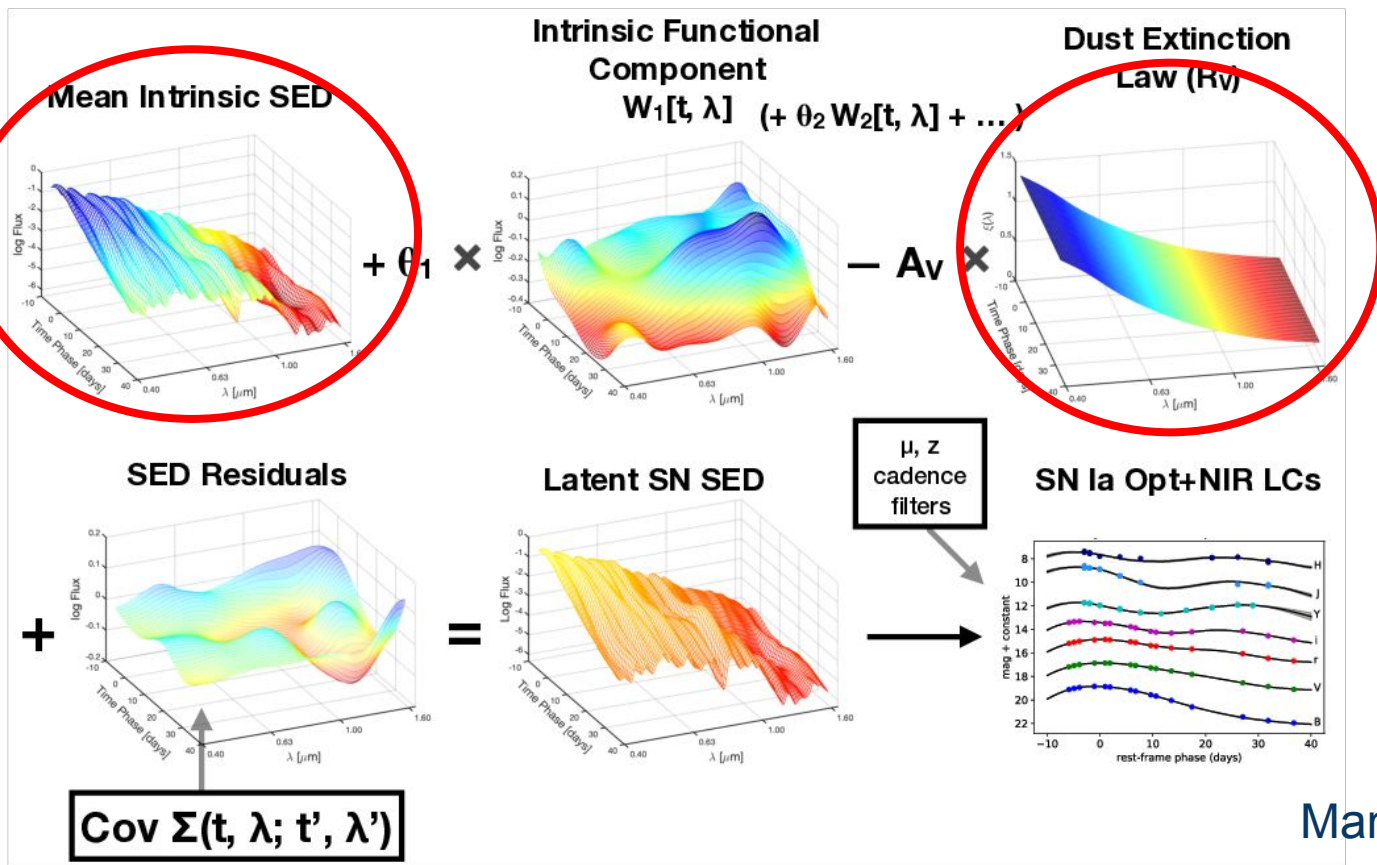
- Hierarchical model enables separate treatments of dust and intrinsic variation
- Uses full light curve in all bands rather than single colour/magnitude pairs
- Can use for both population inference and as a light curve fitter
 - Can fit for distances marginalised over dust distributions

Parameter	Input Value	Estimated Value (Std. Err.)
μ_{R_V} (HM)	3	3.00(0.02)
σ_{R_V} (HM)	0.5	0.63(0.02)
τ_E (HM)	0.125	0.126(0.001)
μ_{R_V} (LM)	2	2.12(0.02)
σ_{R_V} (LM)	0.5	0.41(0.02)
τ_E (LM)	0.08	0.081(0.001)

Grayling & Popovic (2024)

Good recovery of simulated parameters, even with simulations from different model (SALT)

Investigating the Mass Step



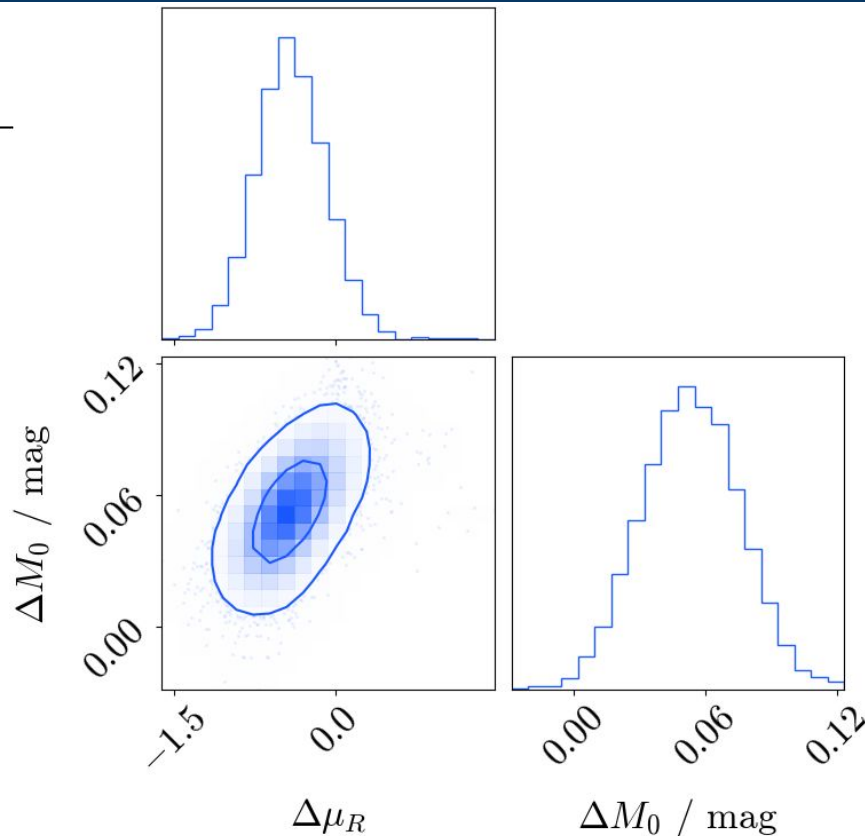
Cause of the Mass Step

Sample	Intrinsic Mass Step / mag
Combined*	0.049 ± 0.019
DES5YR	0.053 ± 0.022

*Combined = DES3YR+Foundation+PS1MD

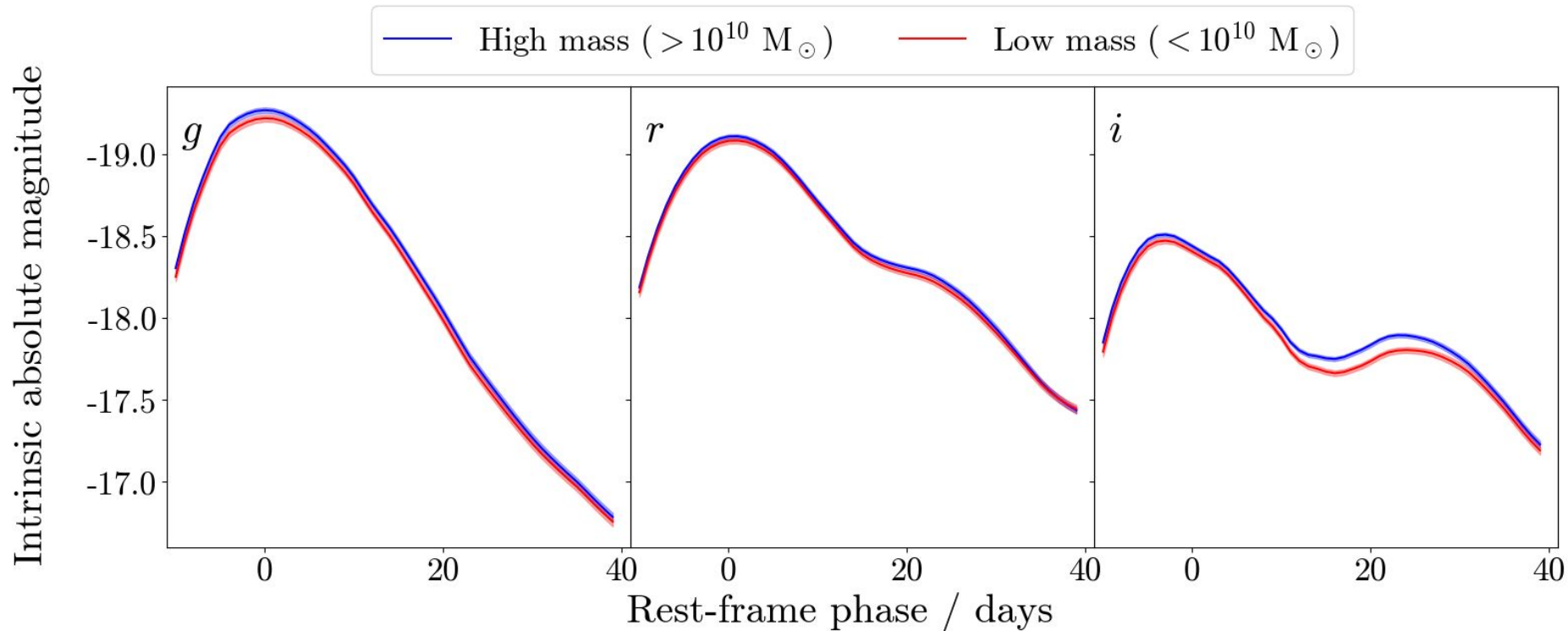
Analysis on volume-limited subsamples

Our results support intrinsic differences and a contribution from dust to explain the mass step



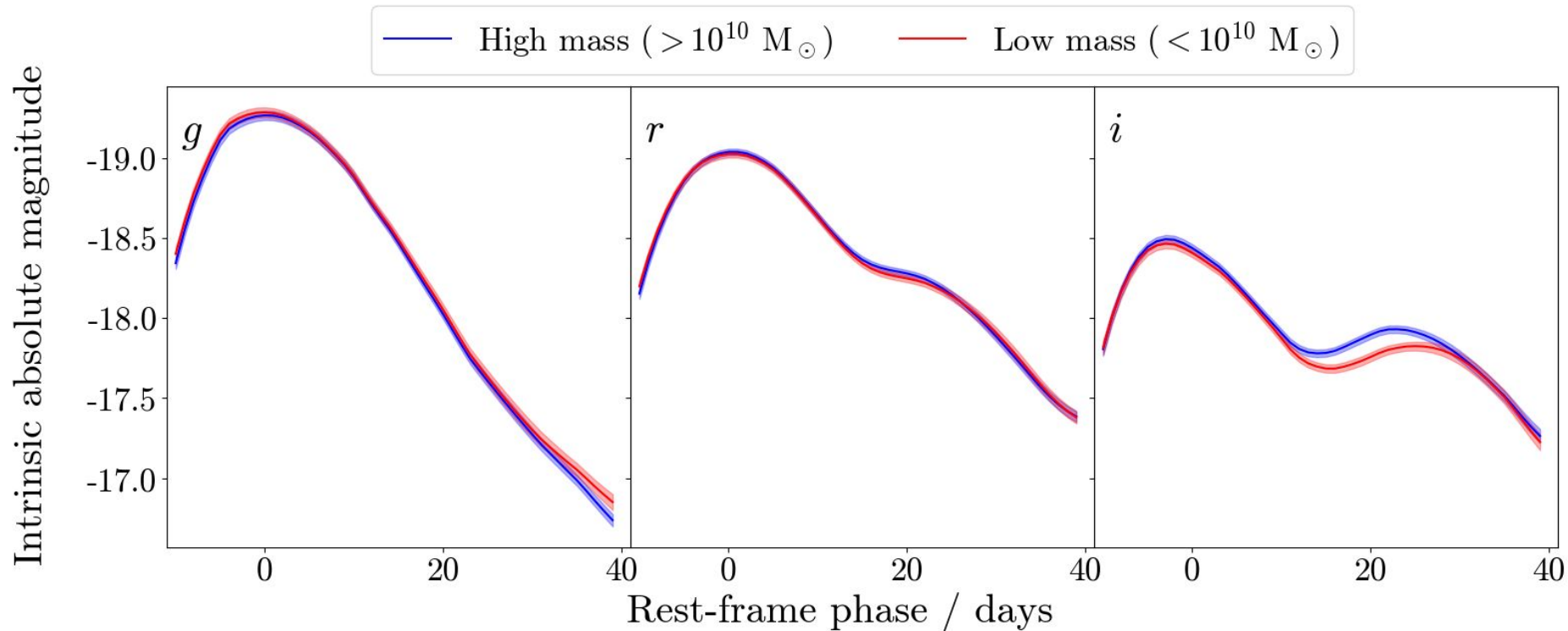
Grayling & Popovic (2024)

DES3YR + Foundation + PS1MD



Grayling+2024

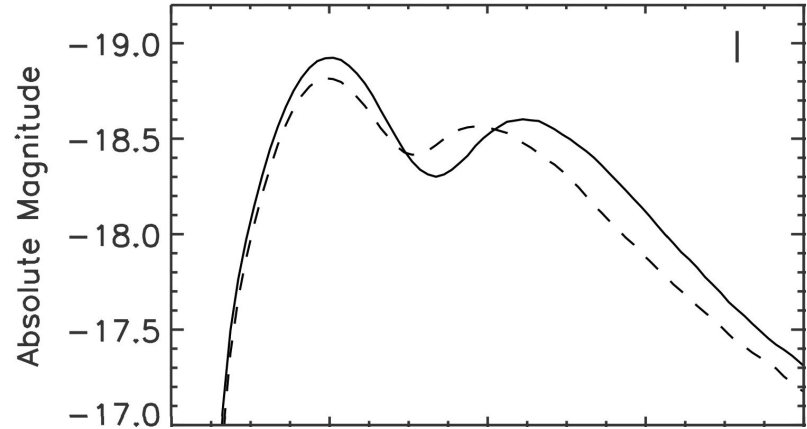
DES5YR



Grayling & Popovic (2024)

What's Causing the i-band Difference?

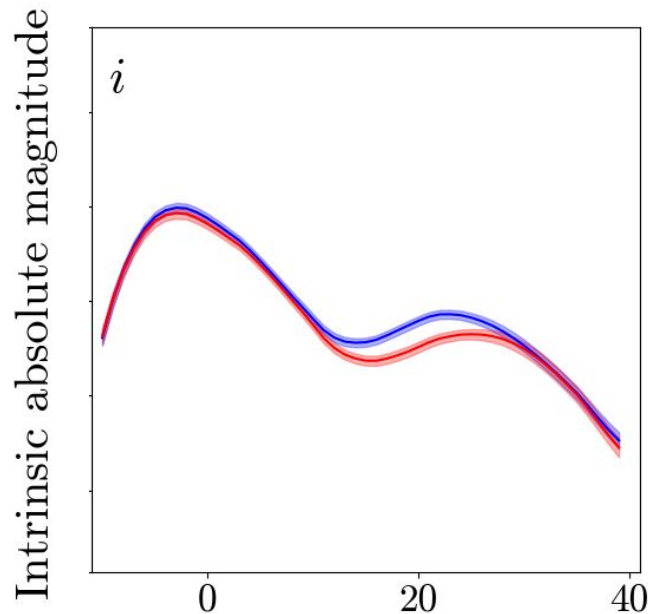
- *Something* is causing an intrinsic difference between SNe Ia in different environments, specifically in i-band.
- Kasen (2006) predicts that metallicity will have stronger impact on secondary maximum than around peak.
- We plan to investigate if this effect persists into NIR wavelengths using the SAINTS sample



From Kasen (2006); dashed line = higher metallicity

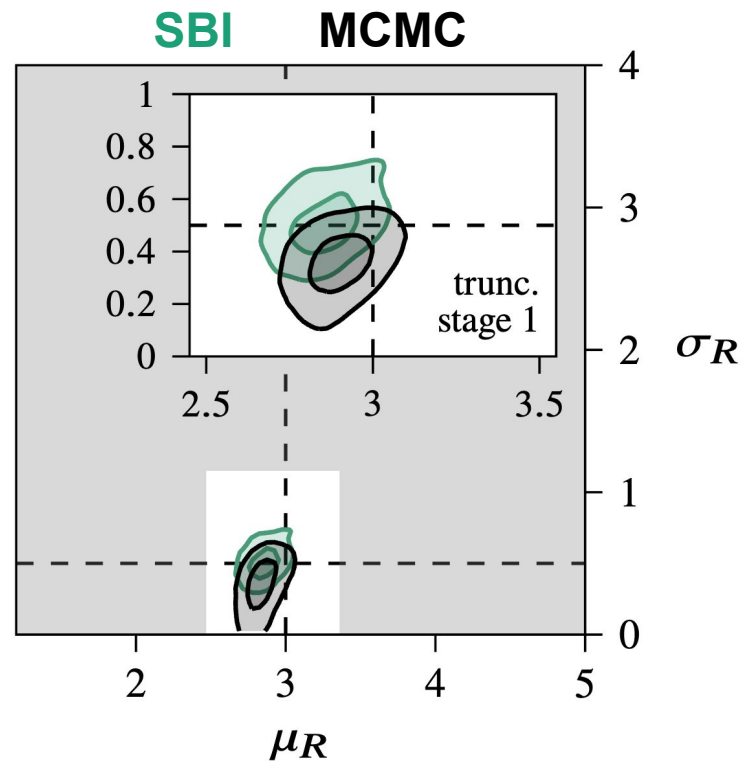
Plans for the Future

- Applying BayeSN to both spectra and photometry
 - Hopefully we can isolate differences in specific spectral features
 - Will provide more physical insight, better understanding of what is driving i-band maximum difference



Plans for the Future

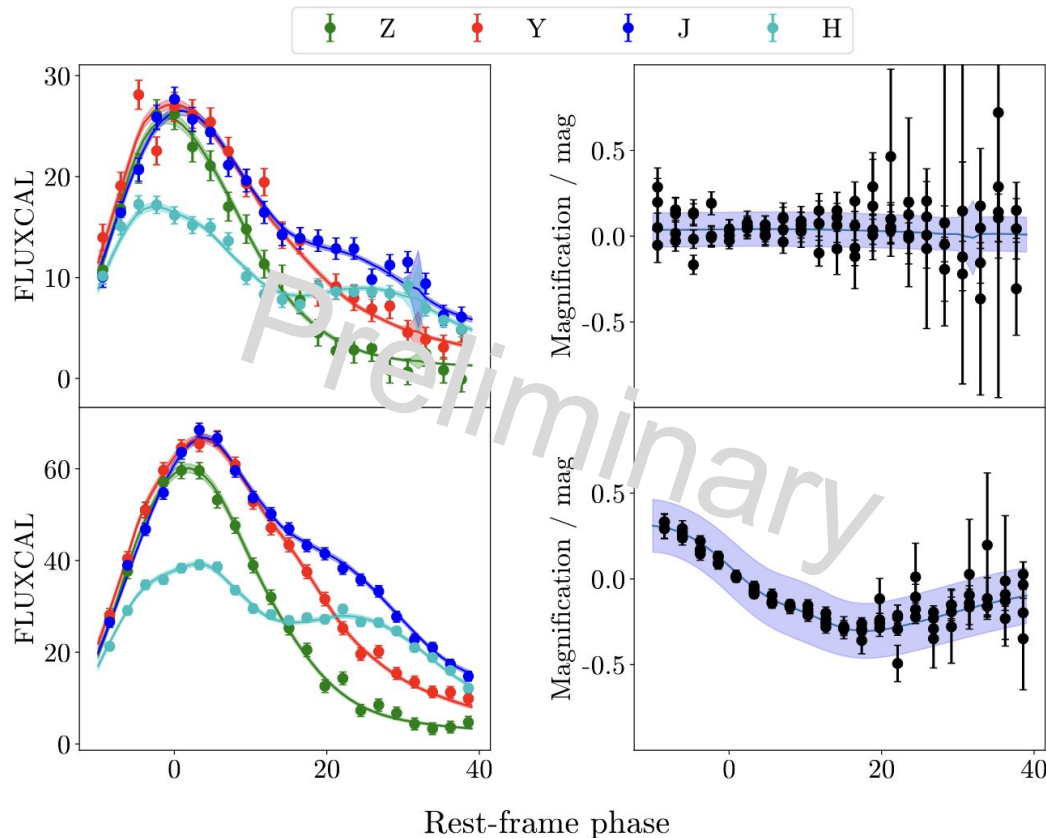
- Cosmology with BayeSN
 - BayeSN added to SNANA, currently testing in standard cosmology pipeline
 - Developing new methods for doing supernova cosmology using Simulation-based Inference (SBI)
 - SBI is the key to inferring cosmological parameters jointly with properties of SNe Ia (e.g. dust)



Karchev, Grayling et al. (2024)

Plans for the Future

- Strongly-lensed SNe
 - New BayeSN-based method for fitting strongly-lensed SNe Ia
 - Allows for deviation from SED template caused by microlensing
 - Very promising results, look for paper soon!



Summary

- BayeSN is a probabilistic SED model for type Ia supernovae, publicly available and incorporated within SNANA
- With BayeSN, we find evidence to support contributions from both dust and intrinsic effects to the mass step
- We find a significant environmental-dependence of the intrinsic properties of SNe Ia, specifically on the i-band secondary maximum
- Lots of ongoing work for BayeSN e.g. training on spectra, moving towards cosmology



MNRAS paper



Grayling+Popovic 2024